4 Phonological and syntactic locality of object clitics in Czech: Evidence from external sandhi

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4.1 Introduction

This chapter focuses on object enclitics in Czech, examining how their phonological integration is influenced by the syntactic and phonological locality of the preceding host. Clitics are generally regarded as deficient forms, and their deficiency is both phonological and syntactic (Zwicky and Pullum, 1983, Cardinaletti and Starke, 1999, Anderson, 2011, Halpern, 2017, Haspelmath, 2023, etc.). On the syntactic side, this deficiency is reflected in their distribution: clitics occupy a limited range of positions in the clause. Some require strict adjacency to a particular syntactic constituent (e.g., Romance clitics are verb adjacent). Others have a fixed clausal position, regardless of what is adjacent to them (e.g., Slavic clitics require second position in the clause). These two distributional patterns are commonly referred to as host-selecting and position-selecting clitics, and they differ in terms of syntactic locality. Host-selecting clitics are, in this sense, syntactically more local, since they require adjacency to a structurally related constituent. Position-selecting clitics, by contrast, are generally expected to be less local, since their distribution is defined by a clausal position rather than by adjacency to a related syntactic constituent.

This distributional contrast can be placed on a general locality scale. At one end of this scale, full words have the weakest requirements on syntactic locality and therefore the widest distribution within the clause. At the other end, affixes have the strongest requirements on syntactic locality and therefore the narrowest distribution. Clitics occupy an intermediate position on this scale, with locality requirements stronger than those of full words but weaker than those of affixes.

However, as argued by Cardinaletti and Starke (1999), among many others, locality differences need not be restricted to contrasts between different forms. In fact, surface forms are often syncretic and may correspond to different degrees of syntactic deficiency, which in turn determine their syntactic distribution. In their approach, syntactic deficiency correlates with structural deficiency, in the sense that the more deficient an item is, the smaller the structure it realises. And more deficient variants have a more limited distribution, whereas less deficient variants have a wider distribution.

Cardinaletti and Starke (1999) illustrate their point with the Geneva French adverb *bien* 'well', which can be deficient and then requires verb adjacency (1a-b), or non-deficient and then displays a freer distribution. In the latter case, *bien* behaves like a full word which can be coordinated, c-modified, or focused (2c).

(1) Distributionally limited adverb in Geneva-French (Cardinaletti and Starke, 1999, p. 98)

a. *Il a bien essuyé la vaisselle.*
b. **Il a essuyé la vaisselle bien.* (verb adjacent *bien* = deficient)
'He has dried the dishes well.'

(2) Distributionally unlimited adverb in Geneva-French (Cardinaletti and Starke, 1999, p. 98)

a. *Il a essuyé la vaisselle bien et rapidement.* (coordinated *bien* = non-deficient)
'He has dried the dishes well and quickly.'
b. *Il a essuyé la vaisselle vraiment bien.* (c-modified *bien* = non-deficient)
'He has dried the dishes really well.'
c. *Il a essuyé la vaisselle bien, pas longuement.* (focused *bien* = non-deficient)
'He has dried the dishes well but at-length.'

Cardinaletti and Starke (1999) further argue that the difference in syntactic locality, which results from the structural deficiency of *bien*, is reflected in its phonological behaviour. Specifically, they show that locality affects the external sandhi process of liaison, as illustrated in (3). In (3a), *bien* occurs in a syntactically local configuration with the verb, and liaison applies. In (3b), *bien* is adjacent to a prepositional adjunct with which it has no syntactic relation, and liaison is not triggered.[1]

(3) Geneva-French liaison (Cardinaletti and Starke, 1999, p. 100)

a. *Je me suis bie**n e**ntendu avec Marie.* [nã] ✓ *bien* shows liaison with the verb
'I got along well with Marie.'
b. *Je m'entends bie**n a**vec Marie.* [∅a] ✗ *bien* does not show liaison with adjunct
'I got along well with Marie.'

Such phonological variability is typically analysed in terms of phonological domains, which delimit the application of phonological processes. Phonological domains to which clitics (CL) attach, either as adjuncts or dependents, include phonological words (PW), phonological phrases (PPh), and intonational phrases (IPh). Table 4.1 summarises the typology of integration patterns as reported in the literature (Klavans, 1985, Selkirk, 1996, Anderson, 2011, Spencer and Luís, 2012, Halpern, 2017).

[1]Much more detailed discussion of standard French liaison can be found in Scheer, 2024.

phonological domain	**(en)clitic integration**
Phonological Word	$[\ldots\ \text{CL}]_{PW}$
	$[[\ldots]_{PW}\ \text{CL}]_{PW}$
Phonological Phrase	$[[\ldots]_{PW}\ \text{CL}]_{PPh}$
	$[[\ldots]_{PPh}\ \text{CL}]_{PPh}$
Intonational Phrase	$[[\ldots]_{PPh}\ \text{CL}]_{IPh}$
	$[[\ldots]_{IPh}\ \text{CL}]_{IPh}$

Table 4.1: Phonological domains of clitic integration

The aim of this study is to investigate integration of object clitics in Czech. As in other Slavic languages, Czech object clitics are traditionally classified as position-selecting (Anderson, 1993, Franks and King, 2000, Zimmerling and Kosta, 2013, Marušič et al., 2024), occupying the second position following the first syntactic constituent of the clause, regardless of its morphosyntactic status. In this canonical second position, they are considered phonologically uniform, meaning that they are assumed to cliticise to the preceding host (Fried, 1994, Toman, 1996, Avgustinova, 1997, Lenertová, 2004). However, as shown in the typological survey in Table 4.2, cliticisation may occur at different phonological levels, which is in turn reflected in phonological behaviour of clitics.

Traditionally, stress has served as the primary diagnostic for different types of cliticisation. However, in Czech, stress is fixed on the initial syllable and is therefore not affected by encliticisation. Moreover, empirical investigation of stress in Czech is challenging, as lexical stress exhibits only weak and inconsistent acoustic correlates (e.g., Volín and Weingartová, 2014, Skarnitzl and Eriksson, 2017). For this reason, we rely on diagnostics other than stress to examine the integration type of object clitics. In particular, we focus on pre-sonorant voicing, which, as an external sandhi process, is expected to vary considerably depending on both syntactic and phonological locality of adjacent lexical items.

The chapter is structured as follows. We first introduce phonological locality and the external sandhi processes that diagnose it. Since there is no systematic typological overview of phonological locality with respect to external sandhi processes, but rather a set of individual studies scattered across the literature, we also bring together the relevant background in this chapter. We then establish the productivity of pre-sonorant voicing in Czech on the basis of a corpus study, in order to determine what can be expected from the process itself and how variable it is in Czech. We then turn to syntactic locality and to the evidence that it conditions phonological locality. We further address production-planning factors that may influence external sandhi, and thus the resulting picture of phonological locality, and we outline how they are controlled for in the experimental design. Finally, we present the production experiment on Czech object clitics and its results.

4.2 Phonological locality of external sandhi

External sandhi processes (such as liaison or pre-sonorant voicing, discussed in the previous section) are typically subject to locality conditions, which constrain the types of word sequences to which they may apply. Broadly speaking, words that occur in a local configuration are candidates for external sandhi, whereas those in a non-local configuration are not. The delineation of locality is language-specific, and consequently, the set of word boundaries across which an external sandhi process applies differs across languages.

Locality conditions on external sandhi have been described in various analyses within the prosodic hierarchy literature (among others Selkirk, 1984, Andersen, 1986, Nespor and Vogel, 1986, Selkirk, 2011). The introductory section above already illustrated how such locality effects have been discussed at clitic boundaries. Since these discussions ultimately rely on a more general notion of phonological locality and on how external sandhi patterns with respect to phonological domains, we now step back from the specific clitic context and summarise the relevant background.

The aim of this section is to provide a more systematic overview of what is currently known about phonological locality and external sandhi, given that the relevant observations are scattered across the literature. We then return to Czech and to the selected external sandhi process, pre-sonorant voicing, by means of a corpus study examining its occurrence before /m/ in general and, more specifically, before the pronominal clitics *mi* 'I.DAT' and *mu* 'he.DAT'.

Prosodic-hierarchy approaches model phonological locality by grouping constituents into hierarchically ordered phonological domains. Prosody is thus organised into tree structures in which some constituents form phonological words (PW), others phonological phrases (PPh), intonational phrases (IPh), or phonological utterances (PU). These four represent the fundamental phonological domains relevant for sandhi processes in general.

When a sandhi process occurs at the boundary between two constituents that form a phonological word, we speak of internal sandhi, which falls outside the scope of this chapter.[2] By contrast, when phonological constituents outgrow the phonological word domain and form a phonological phrase, an intonational phrase, or a phonological utterance, external sandhi emerges, see Table 4.2.

Table 4.2: Phonological domains of external sandhi

phonological domain	**P**honological **Ph**rase	**I**ntonational **Ph**rase	**P**honological **U**tterance
type of boundary	$[...\ \#\ ...]_{PPh}$	$[...\ \#\ ...]_{IPh}$ $[...\ \#\#\ ...]_{IPh}$	$[...\ \#\ ...]_{PU}$ $[...\ \#\#\ ...]_{PU}$ $[...\ \#\#\#\ ...]_{PU}$

The relevant boundaries can be represented on the basis of increasing boundary strength, using different numbers of hash marks. Within a phonological phrase, external sandhi may apply across

[2]In what follows, we are going to discuss and experimentally test pre-sonorant voicing in Czech which is claimed to be an exclusively external sandhi process.

the weakest boundary, the phonological word boundary (#). Within an intonational phrase, it may apply across, on the one hand, the weaker phonological word boundary (#) and, on the other hand, the stronger phonological phrase boundary (##). Finally, within a phonological utterance it may apply across the strongest boundary, the intonational phrase boundary (###), alongside the two weaker ones (# and ##). In each language, external sandhi is defined with respect to one of these domains: it may be defined broadly, with the weakest restrictions on phonological locality, if it is defined for the phonological utterance; it may be defined more narrowly, with stronger locality restrictions, if it is defined for the intonational phrase; or it may be defined most narrowly, with the strongest locality restrictions, if it is defined for the phonological phrase as the smallest domain containing word-external boundaries.

Accordingly, when examining data involving external sandhi processes, (at least) three types of languages can be distinguished, based on the phonological domain with respect to which sandhi is defined, and thus on the strength of the phonological boundary across which external sandhi can apply.

First, there are languages in which external sandhi applies only across the weakest phonological boundary relevant for external sandhi, namely across constituent boundaries within a PPh (#). In these languages, external sandhi is blocked across stronger boundaries (## and ###). Kimatuumbi vowel shortening (Bantu; Odden, 1987) belongs to this type.

(4) Kimatuumbi vowel shortening (examples from Odden, 1987, pp. 14–17)

a.	*naampéi̞ ki̞kól**o**mbe ki̞kúlú*	ki̞kól[o]mbé	#	ki̞kúlú	[o]	✓
	‘I gave him a large shell.’	shell	#	large		
b.	*ki̞kól**oo**mbé ch**aa**púw**aa**ni̞i̞ke*	ki̞kól[oː]mbé	##	ch[aː]púw[aː]ni̞i̞ke	[oː] / [aː]	✗
	‘The shell broke.’	shell	##	broke		
c.	*n**aa**zú̞wi̞ne **aa**kál**aa**nga li̞i̞so*	n[aː]yuwi̞ne	###	[aː]kál[aː]nga	[aː]	✗
	‘Yesterday, I heard that he will fry.’	I-hear	###	he-will-fry		

Second, there are languages in which external sandhi applies not only across constituent boundaries within a PPh (#) but also across stronger constituent boundaries within an IPh (##). Simultaneously, external sandhi is blocked across the strongest constituent boundaries (###) within the PU. This pattern is attested in Slovak and in Cracow–Poznań Polish, both in pre-sonorant and pre-obstruent voicing (Rubach, 1993, Blaho, 2008, Strycharczuk, 2012, Cyran, 2014, Barkanyi and Beňuš, 2015).

(5) Slovak pre-sonorant voicing [3] (examples from Rubach, 1993, Blaho, 2008)

a.	*Je dosť lenivý.*	do**sť**	#	**le**nivý	[zɟl]	✓
	‘He is quite lazy.’	quite	#	lazy		
b.	*Ten chlap môže byť dosť lenivý.*	chla**p**	##	**m**ôže	[bm]	✓
	‘That man may be quite lazy.’	man	##	can		

[3] Examples from the cited literature are adapted to constitute full sentences.

c.	*Aj keď je to dosť lenivý chlap, môže mi pomôcť.* 'Although he is quite a lazy man, he can help me.'	chla**p** man	### ###	**m**ôže can	[pm]	✗

Third, there are languages in which external sandhi applies even across the strongest phonological boundaries (###) within the PU. This type is illustrated by American English flapping (Kilbourn-Ceron et al., 2016) and by RP English linking *r* and intrusive *r* (Vogel, 1986).

(6) RP English intrusive *r* (examples from Vogel, 1986)

a.	*law and order*	l**aw**	#	**a**nd	[ɔːrə]	✓
b.	*Amanda always lies.*	Amand**a**	##	**a**lways	[ərɔː]	✓
c.	*Let's go to Canada. It's cheaper this year.*	Canad**a**	###	**i**t's	[ərɪ]	✓

The typology of the distribution of external sandhi across these phonological domains is summarized in Table 4.3. The application of the process is indicated by a tick in the grey cell, while non-application is indicated by a cross in the white cell.

Table 4.3: Typological overview of the distribution of external sandhi with respect to phonological domains

	across # within Pph	across ## IPh	across ### within PU
Kimatuumbi vowel shortening	✓	✗	✗
Slovak pre-sonorant voicing	✓	✓	✗
RP English intrusive *r*	✓	✓	✓

Table 4.3 summarizes the patterns illustrated by the examples (4)–(6). Kimatuumbi vowel shortening is the most restrictive external sandhi process in terms of its domain of application: it is confined to the phonological phrase and therefore applies only across the weakest boundary (#). Slovak occupies the intermediate position, since pre-sonorant voicing extends to the intonational phrase and can therefore apply across both # and ##. RP English is the least restrictive, since intrusive *r* extends to the phonological utterance and can therefore apply across #, ##, and ###.

From a methodological perspective, prosodic hierarchy theory and research on external sandhi involve an issue. Phonologists define prosodic domains on the basis of external sandhi data and subsequently use these domains to predict the occurrence of external sandhi. As a result, the definition of phonological domains (PPh, IPh and PU) becomes somewhat circular.

To reduce potential circularity, the definition of phonological domains would be strengthened if at least one additional, non-sandhi process also aligns with them. One such process is final devoicing. When we examine the distribution of final devoicing with respect to phonological domains, we find, first, languages that do not show final devoicing at all, apart from utterance-final contexts, where it is phonetically inevitable. This pattern is attested, for example, in (standard) Hungarian[4] (Blaho, 2008, Bárkányi and G. Kiss, 2015) or Breton (Bičanová, 2015, p. 186).

(7) Absence of final devoicing in Hungarian (examples from Bárkányi and G. Kiss, 2015)

a.	*Ezután a smaragd pénzértéke már erősen csökken.*	smara**gd**	#	**p**énzértéke	[gdp]	✗
	'After that, the emerald's monetary value already decreases sharply.'	emerald	#	monetary value		
b.	*A pici gerezd látható szélét penész borítja.*	gere**zd**	##	**l**átható	[zdl]	✗
	'The visible edge of the tiny segment is covered with mold.'	segment	##	visible		
c.	*Lángolva szikrázott a türkiz és a smaragd.*	smara**gd**	###		[kt]~[gd]	✗
	'Blazing, the turqoise and the emerald sparkled.'	emerald	###			

Second, there are languages in which final devoicing occurs across the strongest constituent boundaries within PU (###). It does not apply across weaker constituent boundaries within IPh (##), nor across the weakest constituent boundaries within PPh (#). Slovak and Polish belong to this type (Rubach, 1993, Blaho, 2008, Barkanyi and Beňuš, 2015).

(8) Slovak final devoicing[5] (examples from Rubach, 1993, Blaho, 2008)

a.	*Žaluď je plod duba.*	plo**d**	#	**d**uba	[dd]	✗
	'An acorn is the fruit of the oak.'	fruit	#	oak		
b.	*Už mám tých kliatob dosť.*	kliato**b**	##	**d**osť	[bd]	✗
	'I've had enough of those curses.'	curses	##	enough		
c.	*Už mám dosť tých kliatob, musím ísť preč.*	kliato**b**	###	**m**usím	[pm]	✓
	'I've had enough of those curses. I have to go.'	curses	###	have-to		

Third, we find languages that exhibit final devoicing across strong constituent boundaries (###) within PU and across weaker constituent boundaries (##) within IPh, but not across the weakest constituent boundary (#) within PPh. Czech and Russian represent this pattern (Palková, 1994, Kulikov, 2012, Shrager, 2012, Šimáčková et al., 2012).

[4] Western Transdanubian dialect of Hungarian shows both pre-sonorant voicing and final devoicing. (Bárkányi and Kiss, 2008)

(9) Russian final devoicing[5] (examples from Kulikov, 2012, p. 6)

a.	*Bol pod nami.*	po**d**	#	**n**ami	[dn]	✗
	'He was under us.'	under	#	us		
b.	*Dub upal.*	du**b**	##	**u**pal	[pu]	✓
	'The oak tree fell.'	oak	##	fell		
c.	*Voda byla i pod i nad.*	na**d**	###		[t]	✓
	'Water was both under and over.'	over	###			

The typology of the distribution of final devoicing with respect to phonological domains is summarised in Table 4.4.

Table 4.4: Typological overview of the distribution of final devoicing with respect to phonological domains

	across ## within PPh	across ## IPh	across ### within PU
Czech, Russian	✗	✓	✓
Slovak, Polish	✗	✗	✓
Hungarian	✗	✗	✗

Comparing the distribution of external sandhi procedures (Table 4.3) with the distribution of final devoicing (Table 4.4), we can conclude, that these processes delineate the same phonological domains being mutually in complementary distribution. Thus, the theoretical establishment of the existence of different phonological domains is justified.

If we consider what underlies the delineation of phonological domains in general, it is phonological locality. For external sandhi, locality constitutes a positive requirement. In the most restrictive systems, external sandhi is limited to highly local configurations, namely across the weakest constituent boundaries within the phonological phrase (across # within PPh), where adjacent constituents are maximally local. In other languages, this locality requirement is relaxed, and external sandhi extends to less local contexts, first across stronger constituent boundaries within the intonational phrase (across ## within IPh), and ultimately across the strongest constituent boundaries within the phonological utterance (across ### within PU).

A parallel sensitivity to locality is observed in final devoicing, where locality is negatively marked. In this case, the less local the configuration, the more final devoicing is expected. Accordingly, stronger phonological boundaries, corresponding to the less local phonological configurations, constitute preferred sites for the process, whereas weaker boundaries, corresponding to more local phonological configurations, disfavour it.

At one end of this locality scale, final devoicing is confined to utterance-final position and does not apply across any utterance-internal boundaries, as in Hungarian, (7). In such systems, all constituents contained within the phonological utterance are treated as local. At the opposite end of the

[5] Examples from the cited literature are adapted to constitute full sentences.

locality scale, final devoicing extends to more local configurations within both utterance-internal and intonational-phrase-internal domains, as in Czech, and Russian, and only the phonological phrase is treated as local (recall example (9)).

Depending on how strictly languages define what counts as local, we can posit a cross-linguistic scale of locality requirements, ranging from a strict locality requirement to a relaxed locality requirement, see Table 4.5.

However, it is often difficult to determine which phonological domain two constituents form, and which boundary separates them. Since resolving this issue is not the aim of the present chapter, we adopt a more general and theory-neutral notion of phonological locality. Under this notion, it is assumed only that different types of boundaries may intervene between phonological constituents and that these boundaries are comparable with respect to locality, as can be empirically tested, for example, on the basis of external sandhi and final devoicing.

Table 4.5: Scalar description of external sandhi processes

~~prosodic hierarchy locality~~	~~across # within PP~~	~~across ## within IPh~~	~~across ### within PU~~
theory-neutral locality	**more local**	⇔	**less local**
strict locality requirement	(¬FD) ∧ ES	FD ∧ (¬ES)	FD ∧ (¬ES)
⇕	(¬FD) ∧ ES	(¬FD) ∧ ES	FD ∧ (¬ES)
relaxed locality requirement	(¬FD) ∧ ES	(¬FD) ∧ ES	(¬FD) ∧ ES

The resulting locality scale is not intended to classify phonological objects into strictly defined phonological categories, but rather to compare them relative to one another (e.g., x is more local than y, or y is less local than x).[6] If certain boundaries pattern differently with respect to the processes under consideration, and hence with respect to locality, this provides empirical grounds for treating them as distinct. If, by contrast, they pattern in the same way, they are indistinguishable on the basis of our criteria. This does not necessarily mean that they are identical, but only that no difference can be established using the present method. With respect to such cases, we therefore remain agnostic.

As discussed in this section, external sandhi processes apply within phonological domains of varying size. In the case of Czech, the literature (cited above, or in Section 2.3) suggests that only closely adjacent constituents qualify as phonologically local, which implies a relatively restrictive domain for the application of such processes. If this characterisation is correct, pre-sonorant

[6] This line of reasoning is very similar to Wagner's (2005, p. 135), where he introduces the Boundary Strength Scale.

voicing in Czech should be limited to tightly local configurations, and its non-application should outnumber its application. At the same time, if final devoicing is distributionally complementary to external sandhi processes, its application should outnumber its non-application in comparable contexts.

These predictions can be tested empirically. If pre-sonorant voicing is restricted to narrowly defined phonological domains, we expect lower rates of its application across word boundaries. By contrast, if final devoicing occupies the complementary distribution, higher rates of its application should be observed in the same environments. To evaluate these predictions, we examine the overall distribution of pre-sonorant voicing and final devoicing in naturally occurring data.

In what follows, we therefore present a corpus study examining the distribution of both processes in Czech.

4.2.1 Corpus study on pre-sonorant voicing in Czech

To evaluate the predictions formulated above, we conduct a corpus study examining the distribution of pre-sonorant voicing and final devoicing in Czech. We begin with pre-sonorant voicing triggered by the sonorant /m/ in general, in order to establish its overall distribution and the relative frequency of its application and non-application. The focus on /m/ is methodologically motivated: in the subsequent production experiment, the target items are the clitics *mi* 'I.DAT' and *mu* 'he.DAT'. For this reason, other sonorants are not included in the present analysis.

The second part of the corpus study then narrows the exploration to pre-sonorant voicing occurring specifically before the clitics *mi* and *mu*.

The third part of the study examines Czech final devoicing in parallel with the preceding search. We first consider final devoicing in general before any word beginning with the sonorant /m/, again establishing the relative frequency of its application and non-application, and then focus specifically on contexts in which the following word is the clitic *mi* or *mu*.

The set of voiceless obstruents that constitute potential targets of pre-sonorant voicing in this study comprises /s ʃ x f t c k p/, and the set of voiced obstruents that constitute potential targets of final devoicing comprises /z ʒ ɦ v d ɟ g b/.[7]

As a source of data, we use corpus *Ortofon v3* (Lukeš et al., 2024), which is part of the *Czech National Corpus* and currently represents the largest phonetically annotated corpus of spoken Czech.[8]

To begin with, we conduct a corpus search for pre-sonorant voicing in contexts where word-final voiceless obstruents /s ʃ x f t c k p/ are followed by a word-initial sonorant /m/. In these

[7]Czech alveolar and postalveolar affricates /t͡ʃ t͡s d͡ʒ d͡z/ and fricative /r̝/ were excluded from the corpus study. These segments would be difficult to implement in the subsequent production experiment, as the availability of suitable lexical items containing them is highly limited. In addition, the distribution of voicing within /r̝/ is unstable and depends on the degree of fricativeness, which would introduce an additional source of variability.

[8]The phonetic transcriptions in this corpus are not based on instrumental acoustic analysis but were created manually by trained phoneticians. They, therefore, cannot be treated as fully reliable at the level of fine phonetic detail. Nevertheless, they are sufficient to determine whether pre-sonorant voicing and final devoicing occur at word boundaries in Czech, and whether these processes also appear before the clitics *mi* and *mu*. This provides a solid empirical basis for conducting a subsequent production experiment on these phenomena.

configurations, pre-sonorant voicing may or may not apply: the voiceless obstruents /s ʃ x f t c k p/ may surface as their voiced counterparts /z ʒ ɣ v d ɟ g b/, or retain their voiceless realisation if the process does not apply.

The *Ortofon v3* corpus contains both orthographic and phonetic transcriptions, which allows us to construct queries that simultaneously specify (i) the lexical identity of the final voiceless obstruent (word=".*s"); and (ii) its actual phonetic realisation in the given context (fon=".*s" or fon=".*z"). We therefore search for word pairs in which the first word ends in a lexically voiceless obstruent and the following word begins with /m/ (word="m.*"). An example of such a query for the voicing pair /s–z/ is provided in Table 4.6. The full list of queries is given in the Appendix.

Table 4.6: Queries for pre-sonorant voicing before any word
starting with the phoneme /m/ in *Ortofon v3*

voiceless phoneme	query for lexically voiceless phoneme pronounced as voiceless	query for lexically voiceless phoneme pronounced as voiced
/ s /	[fon=".*s" & word=".*s"][word="m.*"]	[fon=".*z" & word=".*s"][word="m.*"]

Table 4.7: Pre-sonorant voicing before any word
starting with the phoneme /m/ in *Ortofon v3*

voiceless phoneme	pronounced as voiceless (⇒ no pre-sonorant voicing)		pronounced as voiced (⇒ pre-sonorant voicing)		tot.
	abs. freq.	rel. freq. (%)	abs. freq.	rel. freq. (%)	
/ s ʃ x f t c k p /	6420	64.8	3494	35.2	9914

The results of this general search are summarised in Table 4.7. Out of a total of 9914 tokens in which voiceless obstruents precede words beginning with /m/, 3494 instances (35.2 %) involve pre-sonorant voicing, while 6420 instances (64.8 %) retain the voiceless realisation. These data show that non-application outnumbers application, in line with the expectation that in Czech, pre-sonorant voicing is restricted to tightly local configurations.

To test whether pre-sonorant voicing also occurs before the clitics *mu* ‘he.DAT’ and *mi* ‘I.DAT’, we conduct a separate corpus search restricted to these items. Since *mi* and *mu* are traditionally considered to be enclitics and as such are assumed to form a tightly local configuration with the preceding word (Franks and King, 2000, pp. 4–5), pre-sonorant voicing would be expected to apply systematically in this context. The procedure follows the same logic as in the general search, but the following word is limited to the enclitics *mi* and *mu*. A sample query is provided in Table 4.8, and the results of this targeted search are presented in Table 4.9.

Table 4.8: An example of the queries for pre-sonorant voicing before enclitics *mi* 'I.DAT' or *mu* 'he.DAT' in *Ortofon v3*

voiceless phoneme	query for lexically voiceless phoneme pronounced as voiceless	query for lexically voiceless phoneme pronounced as voiced
/ s /	[fon=".*s]" & word=".*s"][word="m[ui]"]	[fon=".*z" & word=".*s"][word="m[ui]"]

Table 4.9: Pre-sonorant voicing before enclitics *mi* 'I.DAT' or *mu* 'he.DAT' in *Ortofon v3*

voiceless phoneme	pronounced as voiceless (⇒ no pre-sonorant voicing)		pronounced as voiced (⇒ pre-sonorant voicing)		tot.
	abs. freq.	rel. freq. (%)	abs. freq.	rel. freq. (%)	
/ s ʃ x f t c k p /	976	66.8	**484**	**33.2**	1460

Out of a total of 1460 tokens in which voiceless obstruents precede the enclitics *mi* or *mu*, 484 instances (33.2 %) involve pre-sonorant voicing, while 976 instances (66.8 %) retain the voiceless realisation. Thus, the rate of application is comparable to that observed in the general search. These data confirm that pre-sonorant voicing occurs before these enclitics, but its application remains variable, despite the fact that they are traditionally expected to form a tightly local configuration with the preceding word.

This variable behaviour of Czech enclitics has been discussed in the literature, for example by Fried (1994). However, the variability described there is limited to cases in which phonological encliticisation is disrupted by the presence of an intonational pause, such as after the insertion of a subordinate clause or a parenthetical expression. The possibility of non-application of encliticisation outside contexts involving a forced phonological boundary is not addressed.

To complement the analysis of pre-sonorant voicing, we also conducted a corpus search for Czech final devoicing. Specifically, we examined cases in which word-final lexically voiced obstruents /z ʒ ɦ v d ɟ g b/ precede a word beginning with the sonorant /m/. In these contexts, final devoicing may or may not apply: the voiced obstruents /z ʒ ɦ v d ɟ g b/ can surface as their voiceless counterparts /s ʃ x f t c k p/ or retain their voiced realisation if the process does not apply. Since final devoicing is assumed to stand in complementary distribution to external sandhi processes, the logic outlined above leads to the following expectation: in contexts where external sandhi is restricted, final devoicing should be favoured. Given that the corpus results for pre-sonorant voicing show that its non-application outnumbers its application in these environments, the complementary process, final devoicing, is expected to show the inverse pattern. Its application should therefore outnumber its non-application.

To test this expectation, we turn to corpus evidence. As in the case of pre-sonorant voicing, the *Ortofon v3* corpus enables us to construct queries that simultaneously specify (i) the lexical identity of the final obstruent (word=".*z") and (ii) its actual phonetic realisation (fon=".*s" or fon=".*z"). We therefore searched for word pairs in which the first word ends in a lexically voiced obstruent and the following word begins with /m/ (word="m.*"), while distinguishing between contexts where the final obstruent was realised as voiceless (indicating final devoicing) and those where it remained voiced. An example of such queries is provided in Table 4.10, and the results of this general search are summarized in Table 4.11.

Table 4.10: An example of the queries for final devoicing before any word starting with the phoneme /m/ in *Ortofon v3*

voiced phoneme	query for lexically voiced phoneme pronounced as voiceless	query for lexically voiced phoneme pronounced as voiced
/ z /	[fon=".*s" & word=".*z"][word="m.*"]	[fon=".*z" & word=".*z"][word="m.*"]

Table 4.11: Final devoicing before any word starting with the phoneme /m/ in *Ortofon v3*

voiced phoneme	pronounced as voiceless (⇒ final devoicing)		pronounced as voiced (⇒ no final devoicing)		tot.
	abs. freq.	rel. freq. (%)	abs. freq.	rel. freq. (%)	
/ z ʒ ɦ v d ɟ g b /	**1928**	**53.2**	1695	46.8	3623

The data show that final devoicing applies before words beginning with /m/, but not categorically. Out of a total of 3623 relevant tokens, 1928 instances (53.2 %) involved devoicing, while 1695 instances (46.8 %) retained voicing. This confirms that final devoicing across word boundaries before /m/ is attested in Czech, yet variable.

At the same time, the overall number of relevant tokens remains limited, and the corpus alone does not provide sufficient data to evaluate the role of specific conditioning factors in a controlled manner. This limitation motivates the need for additional data collected under experimental conditions. The production experiment presented in Section 4.5 is designed to partially fill this empirical gap.

In addition to the general context in which a word-final voiced obstruent precedes any word beginning with /m/, we also examine cases in which this second word is an enclitic (*mi* or *mu*). As noted above, the literature assumes that enclitics form a tightly local configuration with the preceding word. In such tightly local contexts, external sandhi should be favoured. If the complementarity holds, final devoicing should therefore be disfavoured and its non-application should be more frequent. To test this prediction, we conducted a separate corpus search restricted to the enclitics *mi* and *mu*. A sample query for this search is provided in Table 4.12, and the results of this targeted search are summarized in Table 4.29.

Table 4.12: An example of the queries for final devoicing before enclitics *mu* 'he.DAT' or *mi* 'I.DAT' in *Ortofon v3*

voiced phoneme	query for lexically voiced phoneme pronounced as voiceless	query for lexically voiced phoneme pronounced as voiced
/ z /	[fon=".*s" & word=".*z"][word="m[ui]"]	[fon=".*z" & word=".*z"][word="m[ui]"]

Table 4.13: Final devoicing before any word starting with the phoneme /m/ in *Ortofon v3*

voiced phoneme	pronounced as voiceless (⇒ final devoicing)		pronounced as voiced (⇒ no final devoicing)		tot.
	abs. freq.	rel. freq. (%)	abs. freq.	rel. freq. (%)	
/ z ʒ ɦ v d ɟ g b /	**325**	**60.0**	217	40.0	542

The results indicate that final devoicing also occurs before the enclitics *mi* and *mu*. Out of 542 relevant tokens, 325 instances (60.0 %) involve devoicing, while 217 instances (40.0 %) retain voicing. In this respect, the findings are consistent with the complementary relation between final devoicing and pre-sonorant voicing, as the application of final devoicing outnumbers its non-application. At the same time, however, these results run counter to the established view that Czech enclitics display non-variable tightly local behaviour in which final devoicing should be disfavoured.

Taken together, the corpus results show that in Czech both pre-sonorant voicing and final devoicing occur in sequences in which word-final obstruents are followed by words beginning with the sonorant /m/, including cases where this second word is the enclitic *mi* or *mu*. At the same time, neither phonological process applies uniformly in these environments. The variable behaviour of the enclitics *mi* and *mu* further suggests that they do not consistently occur in a single phonological configuration, but rather in configurations reflecting different degrees of phonological locality. The observed distribution, therefore, indicates that additional factors shape the behaviour of these processes in Czech, a pattern that has not been explicitly reflected in the existing literature.

However, the *Ortofon v3* corpus is relatively limited in size and uneven with respect to structural contexts to capture this. For this reason, we proceed by collecting our own linguistic material from native speakers of Czech in a controlled production experiment. This design enables us to systematically manipulate the relevant structural configurations while controlling for other factors that might obscure their effect.

Further, the variability in the data raises a theoretical question: which factors, beyond the phonological locality itself, shape the distribution of these processes? Phonological realisation does not operate in isolation. It plausibly responds either to properties of the grammar that precede it, such as (morpho)syntactic organisation, or to properties that follow it, namely factors related to speech production.

In what follows, we focus on syntactic conditioning of the external sandhi processes (and final devoicing) as a grammatical factor (Section 4.3), and on speech production conditioning of these processes as an extra-grammatical factor (Section 4.4). In addition to phonological locality, we therefore distinguish two further types of locality: syntactic locality and the locality of production planning.

Turning first to the grammatical factor mentioned above, the relationship between phonology and (morpho)syntax has long been a central topic in the literature and continues to be the subject of competing analyses. With respect to (morpho)syntactic factors, the linguistic field divides into two major positions. One view holds that phonological and morphosyntactic locality are not always isomorphic, a position supported by a wide range of so-called bracketing paradoxes, prominently discussed in Nespor and Vogel (1986). The opposing view maintains that phonological locality mirrors morphosyntactic structure, proposing a systematic mapping from morphosyntactic constituents onto phonological domains. This position is defended, for instance, by Kaisse (1985), Odden (1987) and Newell and Piggott (2014).

In the following section, we adopt the second view, namely that phonological locality can be mapped from (morpho)syntactic structure. There is good reason to pursue this route in the case of clitics, which are inherently defined at the intersection of (morpho)syntax and phonology. Their behaviour is constrained within these grammatical domains: (morpho)syntactically, clitics are restricted to specific positions within the clause; phonologically, they interact with adjacent constituents located either to their left (enclitics) or to their right (proclitics). This interaction, however, may be blocked if a strong phonological boundary intervenes (e.g., Fried, 1994). The question we address is therefore whether particular syntactic configurations favour or disfavour the application of sandhi processes at clitic boundaries in terms of locality.

This line of inquiry is not without precedent. In the Czech literature, variability of sandhi processes at proclitic boundaries has been observed with prepositions, where syntactic locality has been shown to condition whether such processes apply (Ondráčková, 1954, Zeman, 1983, Palková, 1994, p. 339, Skarnitzl, 2014). By contrast, no comparable discussion exists for enclitics (except for several intuitive remarks in Trávníček, 1959). Our goal is therefore to test the enclitic context.[9]

Since we assume that syntactic locality determines phonological locality, and thereby the application of external sandhi processes, we first review how this relationship has been treated in the literature and then turn to the case of Czech.

4.3 Syntactic locality of external sandhi

First, we illustrate how syntactic locality determines the application of external sandhi. We build on the observation that phonological processes across word boundaries are inherently variable, with one of the key factors in this variability being the syntactic locality between adjacent words (Kaisse, 1985, Klavans, 1985, Kaisse and Zwicky, 1987, Embick, 2010, Marantz, 2013, Newell and Piggott, 2014, Scheer, 2024). In what follows, we present two examples from English and Russian that demonstrate how syntactic locality influences the application of external sandhi both at boundaries between two full words and at boundaries between a clitic and a full word.

[9] For more details see Section 4.3.1 Czech pre-sonorant voicing in the clitic context.

We begin with an example involving a boundary between two full words. Bailey (2019) focuses on the word-final /ng/ clusters in English, since they exhibit variable pronunciation, either as [ŋg] or [ŋ]. To detect the factors of this variability, Bailey conducts a production experiment with native speakers of North-Western English. Among other things, the results show that the greater the syntactic distance between an /ng/-final word and the following consonant-initial word, the more likely the cluster is to be pronounced in its full form as [ŋg] since the shape of the following word cannot induce the simpler [ŋ] form. In other words, the cluster is more likely to be realised as [ŋg] in less local configurations (such as (10c) or (10d)) than in more local configurations (such as (10a) or (10b)), where the /ng/-final word is situated within one syntactic, and thus one phonological domain. The tendency for cluster preservation was even stronger when the phonological boundary between the target items was reinforced by a perceivable intonational pause.[10]

(10)

a.	*She was given the wrong amount.* [the wrong amount]$_{NP}$ wro**ng** # amount → [ŋ]	→ more local: they form a phonological domain for cluster reduction (i.e., weak phonological boundary favours cluster reduction)
b.	*She gave the ring a quick polish.* [gave[the ring]$_{NP}$[a quick polish]$_{NP}$]$_{VP}$ ri**ng** ## a quick → [ŋ]	→ more local: they form a phonological domain for cluster reduction (i.e., weak phonological boundary favours cluster reduction)
c.	*"It's a traditional thing," Patricia said.* [[it's a traditional thing]$_{CP}$ Patricia said]$_{CP}$ thi**ng** ### Patricia → [ŋg]	→ less local: they do NOT form a phonological domain for cluster reduction (i.e., strong phonological boundary favours cluster preservation)
d.	*The drink was surprisingly strong.* [the drink was surprisingly strong]$_{CP}$ stro**ng** ### → [ŋg]	→ less local: no following material is available for the formation of a shared phonological domain (i.e., strong phonological boundary favours cluster preservation)

Another example of the influence of syntactic structure on external sandhi is the behaviour of prepositions in Russian. Gribanova and Blumenfeld (2013) illustrate external sandhi effects in prepositional structures through two processes: stress placement and vowel epenthesis. There are

[10]For each configuration type, (10a-d), there were also examples in which the following word began with either a vowel or a consonant. The four examples presented are illustrative and are intended to represent typical cases drawn from the actual dataset. For instance, a parallel sentence with a consonant-initial word for type (10a) could be *She was given the wrong book*. However, Bailey (2019) does not provide the full list of stimuli used in the experiment, so we were not able to include them here.

two things one has to know about Russian to understand these processes: firstly, Russian prepositions are proclitics that form single stress-bearing domains with the following word, secondly, in Russian, lexical stress is not fixed but variable, and this variability is also reflected at the proclitic boundary of the preposition. According to Gribanova and Blumenfeld (2013), proclitic prepositions such as *pod* 'under' and *do* 'until' follow either a iambic pattern, in which stress is retained on the noun, or a trochaic pattern, in which the preposition retracts stress from the noun. They argue that this iambic–trochaic variation is influenced by syntactic configuration. The trochaic pattern involves a close syntactic relationship between the preposition and the noun, which is, among other things, also reflected in the idiomatic meaning of the prepositional phrase (i.e., local syntactic configuration implies more local phonological configuration). The iambic pattern, on the other hand, corresponds to some extent of syntactic and semantic independence, which is also reflected in the compositional meaning of these phrases (i.e., less local syntactic configuration). Compare the examples (11a-b) with (11c-d) below.

(11)	a.	*pod goru* 'downhill' [PODgoru]	→ more local: they form a phonological domain for stress assignment, the preposition
	b.	*do smerti* 'extremely' [DOsmerti]	receives stress (i.e., weak phonological boundary favours cluster reduction)
	c.	*pod goru* 'under the hill' pod [GOru]	→ less local: they do NOT form a phonological domain for stress assignment,
	d.	*do smerti* 'until death' do [SMERti]	the preposition remains stressless (i.e., strong phonological boundary disfavours stress-shift)

Similarly, Russian consonantal prepositions such as *k* 'to' exhibit vowel epenthesis, and the *k*~*ko* alternation is sensitive to syntactic configuration. Vowel epenthesis is obligatorily triggered when the nominal constituent, such as the personal pronoun *mne* 'I.DAT', is a direct complement of the preposition, (12a). By contrast, the preposition appears without the epenthetic vowel when an intervening syntactic constituent (e.g., a complex attribute) is present, as in (12b). Compare the examples in (12a) and (12b) below.[11]

[11] Each example contains five components: (i) the example sentence with its translation: *Ona prišla ko mne.* 'She came to me.', (ii) the indication of syntactic locality within the relevant stretch of the example: [k mne]$_{PP}$, (iii) the indication of phonological locality within the same stretch: **k** # **mn**e (more local) → [**komn**], (iv) the gloss of that stretch: to me.DAT, and (v) a specification of whether the configuration represents a more local or less local context and which process is favoured or disfavoured accordingly: they form a phonological domain for vocalisation of *k* to *ko* (i.e., weak phonological boundary favours vocalisation).

(12) a. *Ona přišla ko mne.*
‘She came to me.’
[k mne]$_{PP}$
to me.DAT
k # **mn**e → [**komn**]

→ more local:
they form a phonological domain
for vocalisation of *k* to *ko*
(i.e., weak phonological boundary
favours vocalisation)

b. *Ona přišla k mne neizvestnomu čeloveku.*
‘She came to a person unknown to me.’
[k [[mne neizvestnomu]$_{AP}$ čelověku]$_{NP}$]$_{PP}$
to me.DAT unknown.DAT person.DAT
k ## **mn**e → [**kmn**]

→ less local:
they do NOT form a phonological
domain for vocalisation of *k* to *ko*
(i.e., strong phonological boundary
disfavours vocalisation)

Based on these observations, we expect Czech clitics to show similar phonological behaviour, conditioned by their syntactic configuration.

Similarly to Russian, Czech has proclitic prepositions which, although described unsystematically with respect to this property, are documented to exhibit the same syntactic conditioning of phonological processes (Vachek, 1968, Zeman, 1983, Fried, 1999, Skarnitzl, 2011, 2014, Krčmová, 2017). Furthermore, Czech proclitic prepositions display the same external sandhi processes as Russian, namely pre-obstruent voicing assimilation (e.g., /k z/*emle* → [gz]*emle* ‘to the ground’; Kulikov, 2012, p. 426) and the vocalisation of consonant-final prepositions (already illustrated in (12a). The corresponding Czech patterns are shown in ((13a-d); adapted by the authors for this chapter from Kučera, 1961, Zeman, 1983, Lundová, 2018).

(13) a. *Tahle cesta vede k Boru.*
‘This road leads to Bor.’
[k Boru]$_{PP}$
to Bor.ACC
k # **b**oru → [**gb**]

→ more local:
they form a phonological domain
for pre-obstruent voicing assimilation
(i.e., weak phonological boundary
favours voicing assimilation)

b. *Donesl to ke mně.*
‘He brought it to me.’
[k mně]$_{PP}$
to me.DAT
k # **mn**ě → [**kemɲ**]

→ more local:
they form a phonological domain
for vocalisation of *k* to *ke*
(i.e., weak phonological boundary
favours vocalisation of *k* to *ke*)

c. *Cesta vede k Borem vlastněným pozemkům.*
‘This road leads to land owned by Bor.’
[k [[[Borem vlastněným]$_{AP}$ pozemkům]$_{NP}$]$_{PP}$
to Bor.INST owned.DAT lands.DAT
k ## **b**orem → [**kb**]

→ less local:
they do NOT form a phonological
domain for pre-obstruent voicing
assimilation
(i.e., strong phonological boundary
disfavours voicing assimilation)

d.	*Donesl to <u>k mně</u> neznámému člověku.* 'He brought it to the person unknown to me.' [k[[mně neznámému]$_{AP}$ člověku]$_{NP}$]$_{PP}$ to me.DAT unknown.DAT person.DAT <u>**k**</u> ## <u>**mn**</u>ě → [**kmɲ**]	→ less local: they do NOT form a phonological domain for vocalisation of *k* to *ke* (i.e., strong phonological boundary disfavours vocalisation)

Both sandhi processes in (13a-d) are claimed to be sensitive to syntactic locality. Thus, in (13a-b), we observe pre-obstruent voicing assimilation and vocalisation of prepositions in the more local syntactic configuration, whereas examples (13c-d) display the absence of pre-obstruent voicing assimilation and vocalisation of prepositions in the less local syntactic configuration, where pronominal and adjectival complements intervene between the prepositions and the nouns they modify. Thus, it has been shown that Czech proclitics show sensitivity to syntactic locality, and based on this, we also assume an analogous behaviour for Czech enclitics.

This has already been hypothesized in an experimental study by Poĺomská and Ziková (2024) on the Czech pronominal enclitic *ho* 'him.ACC', which was tested as a trigger of an external sandhi process, namely pre-obstruent voicing assimilation. The results confirm syntactic locality as a factor in the phonological behaviour of the pronominal enclitic: constituents that are syntactically more local to the enclitic (verbs) show a statistically different proportion of pre-obstruent voicing assimilation than constituents that are syntactically less local to the enclitic (adverbs), with the former showing a much stronger tendency for pre-obstruent voicing assimilation than the latter.

These findings correspond to the conclusions of Kaisse (1985), who likewise compares processes at the boundary between a clitic and its phonological host with external sandhi processes at the boundary between two full words. With respect to the behaviour of external sandhi at the boundary between a clitic and a more local vs a less local host, she formulates a clear implicational generalisation:

(14) [...] *if a clitic undergoes a given phonological process with a non-host* [i.e., a syntactically less local constituent], *it may also undergo that process with its host* [i.e., a syntactically more local constituent]. (Kaisse, cited in Klavans, 1985, p. 107)

This implicational generalisation provides a different way of stating the locality asymmetry that we have illustrated visually in Table 4.5.

Our intention to test different degrees of locality at the word-enclitic boundary by means of external sandhi is therefore theoretically well motivated. Before doing so, however, it is necessary to review what is already known about the phenomenon in Czech.

In the next section, we summarise all available sources known to us on the topic and outline the assumptions that guide our investigation.

4.3.1 Czech pre-sonorant voicing in the clitic context

In Czech, pre-sonorant voicing constitutes one of the external sandhi processes. In the relevant phonological literature, however, it is mentioned only rarely, and its sensitivity to syntactic configuration is discussed even more rarely. Moreover, its interconnection with the enclitic context is entirely absent.

Fragmentary mentions of pre-sonorant voicing in Czech appear primarily in discussions of proclitic prepositions and their interaction with the nouns they modify (Ondráčková, 1954, Zeman, 1983, Krčmová, 2009, p. 33–35, Skarnitzl, 2011, 2014). This syntactic configuration gives rise to a number of processes discussed above, including pre-obstruent voicing assimilation, vocalisation of prepositions, and various patterns of stress assignment. Pre-sonorant voicing is occasionally mentioned in this context as well.

Examples that should exhibit different behaviour with respect to pre-sonorant voicing precisely due to differences in syntactic locality are provided below in (15a-b). This behaviour has been documented in the corpus study by Ptak (2022), but its consistency has not been established due to insufficient data.

(15)	a.	*Šel k městu.* 'He walked to the city.' [k městu]$_{PP}$ to city.ACC **k** # **m**ěstu → [**gm**]	→ more local: they form a phonological domain for pre-sonorant voicing (i.e., weak phonological boundary should favour this process)
	b.	*Šel k městem vlastněným pozemkům.* 'He walked to the city-owned land' [k[[městem vlastněným]$_{AP}$ pozemkům]$_{NP}$]$_{PP}$ to city.INSTR owned.DAT lands.DAT **k** ## **m**ěstem → [**km**]	→ less local: they should NOT form a phonological domain for pre-sonorant voicing (i.e., strong phonological boundary should disfavour this process

In example (15a), the proclitic preposition *k* 'to' occurs adjacent to the dative form of the noun *městu* 'city', with which it is syntactically related, and pre-sonorant voicing is therefore applied. This syntactic configuration is thus considered more local. By contrast, in example (15b), the instrumental form of the noun *městem* 'city' is not syntactically related to the preposition, but to the intervening adjective, and pre-sonorant voicing is therefore not expected to occur between *k* and *městem*. This configuration is considered less local.

The generalisation illustrated by these examples is thus that pre-sonorant voicing applies in more local syntactic contexts and is avoided in less local ones. Our aim is to confirm or refute a parallel locality effect in the case of Czech pronominal enclitics.

Similar to the earlier mentioned proclitic prepositions, Czech pronominal enclitics may be situated in the more or less local syntactic configurations. In case they are placed after verbs, we assume that they are in the local configuration since there is a syntactic relationship between the verb and its pronominal argument, implying that the pre-sonorant voicing may be applied (see (16a)).[12] On the other, in case the pronominal enclitics are placed after the adverbs, there is no syntactic relationship between the adverb and the pronominal argument. This corresponds to the

[12] Avgustinova and Oliva (2016) mention a closer relationship between verbs and enclitics, though not in terms of locality. Rather, they relate this tendency to properties of clitic systems in which clitics are obligatorily attached to a verbal host – comparably to obligatorily verb-adjacent clitics in Romance languages. Trávníček (1959) also mentions

non-local configuration, which subsequently implies that pre-sonorant voicing should be avoided (see (16b)).

(16) a. *Přines mu to do kanceláře.* → more local:
'Bring it to his office.' they should form a phonological
[přines mu]$_{VP}$ domain for pre-sonorant voicing
bring.IMP.2SG he.DAT (i.e., weak phonological boundary
přine**s** # **m**u → [**zm**] should favour this process)

b. *Dnes mu obváže koleno.* → less local:
'Today, he will bandage his knee.' they should NOT form a phonological
[dnes [mu [obváže koleno]]]$_{CP}$ domain for pre-sonorant voicing
today he.DAT (i.e., strong phonological boundary
dne**s** ## **m**u → [**sm**] should disfavour this process)

The behaviour of the examples in (16a-b) remains hypothetical. To our knowledge, these examples have not been discussed in connection with pre-sonorant voicing and locality effects. Our aim is therefore to examine this largely unexplored area and to provide new data.

In addition to syntactic effects on phonological locality, there is another source of variability coming from outside the grammar, namely speech production planning (Wagner, 2012, Kilbourn-Ceron, 2017, Kilbourn-Ceron et al., 2020). The next section focuses on this third component underlying the distribution of external sandhi processes.

4.4 Locality of production planning

The previous sections have shown that pre-sonorant voicing is conditioned by phonological and syntactic locality. Under this view, once the relevant phonological and syntactic configurations are specified, the grammatical system yields categorical predictions: in a given structural configuration, the process is either licensed or not.

Nevertheless, empirical data show that external sandhi processes, including pre-sonorant voicing, remain variable even in contexts that are structurally maximally local and where the process should apply. At the same time, variability is also observed in contexts where the structural configuration should disfavour the process.

If phonological and syntactic locality fully determine the grammatical conditioning of the process, such residual variability cannot be attributed to these factors. The source of this variation must therefore lie outside the grammatical system.

A possible source of such variability is speech production planning. During online speech processing, phonological encoding proceeds in planning units that may correspond to syntactic or phonological constituents, but need not do so. These units may be either smaller or larger than the corresponding syntactic or phonological constituents. As a consequence, syntactically or

his intuitive assumption about closer relationship between verbs and their objects and its subsequent phonological reflections.

phonologically local constituents may be processed separately, while syntactically or phonologically non-local constituents may be processed together within a single production planning domain.

The theory of speech production includes the Locality of Production Planning Hypothesis (Wagner, 2012, Tanner et al., 2015, Kilbourn-Ceron et al., 2016). This hypothesis proposes that constraints on online speech production planning play a role in explaining external sandhi patterns. The core idea is the following: given a pair of adjacent words, the pronunciation of the first word (w1) cannot be influenced by the initial segment of the second word (w2) if w2 does not enter the same production planning window as w1. If w1 is encoded without w2 being part of that window, cross-word interaction is blocked and external sandhi does not apply.

The size of the production planning window therefore underlies the observed variability. This explanation presupposes that the size of the window is not fixed, but may expand or contract depending on various factors, including the complexity of the syntactic or phonological constituent being planned (Ferreira, 1991) and other cognitive factors, such as working memory load (Wagner et al., 2010, Slevc, 2011).

Having discussed the two main grammatical conditioning factors (phonological and syntactic locality), the key issue now concerns the extra-grammatical conditions under which two adjacent elements are treated as local within a single production planning domain. Identifying these factors is essential, as it enables us to control for them in the experimental design and thereby isolate the genuinely grammatical sources of variability.

In what follows, we therefore list four production-related factors that may influence whether w1 (an obstruent-final word) and w2 (a following sonorant-initial word or clitic) are encoded within the same production planning domain: (i) lexical frequency of w1 and w2; (ii) conditional probability of w2 given w1; (iii) duration of the relevant words; and (iv) speech rate and (v) individually set working memory load (Wagner, 2012, Kilbourn-Ceron, 2017).

In the following subsection, we describe the production-planning factors mentioned above and how our experimental setting tries to control for them.

4.4.1 How to control for the production-planning factors

Lexical frequency has been argued to interact with production planning. High-frequency words are retrieved more rapidly, and their phonological forms become available earlier during encoding. Under this view, higher lexical frequency of either w1 or w2 increases the likelihood that both words are encoded within the same production planning domain, thereby facilitating cross-word interaction and the application of external sandhi (cf. Kilbourn-Ceron, 2017; Dunagan and Renwick, 2021).

In the subsequent study, w2 consists of a pronominal enclitic (*mi* or *mu*), which are functional elements and therefore inherently high-frequency lexical items. Their high frequency predicts early availability during encoding and thus favours joint parsing with the preceding word. The w1 items are verbs or adverbs. These shall be selected so as to represent neutral, non-specialized vocabulary, avoiding expressive, technical, dialectal, or otherwise marked items. Given the lexical profile of the selected items, both adverb-CL and verb-CL configurations from the upcoming experiment are expected to favour joint encoding within a single production planning domain. Under this assumption, pre-sonorant voicing should be favoured, while final devoicing should be disfavoured with respect to the lexical frequency of both w1 and w2.

Another factor that may influence joint encoding is the conditional probability of w2 given w1, that is, the extent to which the occurrence of w2 is statistically associated with a particular preceding word. In the case of Czech pronominal enclitics, however, this factor is structurally neutralised. Czech pronominal enclitics are uniformly placed in the second position of the clause, regardless of the lexical identity or category of the first constituent. As Wackernagel clitics, they systematically occupy the position immediately following the initial clausal element. Their occurrence is therefore determined by clause structure rather than by lexical selection. For this reason, the conditional probability of w2 given w1 is not expected to vary systematically across different w1 items in our design. Consequently, with respect to this factor, the inclusion of w1 and w2 within a single production planning domain should not depend on which particular word serves as w1.

Word duration constitutes another factor that may affect production-based locality. Longer words require more time to encode and may therefore increase the likelihood that adjacent items are planned in separate production planning domains. In Czech, the most frequent lexical word lengths are monosyllabic, bisyllabic, and trisyllabic (Bičan, 2015, p. 100). From the perspective of production planning, sequences consisting of two short words, such as two monosyllabic words or a bisyllabic word followed by a monosyllabic word, should not in themselves exceed the temporal limits of a single production planning domain in Czech.

To minimize variability related to duration, we thus control phonological length in the experimental design, so that the resulting sequences fit within the trisyllabic template commonly attested in Czech. This restriction reduces the likelihood that separation into distinct production domains is driven by word length/duration rather than by the grammatical factors under investigation.

Finally, speech rate may influence the size of the production planning domain. There is evidence suggesting a correlation between higher speech rate and larger planning domains (Wagner et al., 2010). To account for this factor, participants were instructed to speak at their own comfortable pace, avoiding both deliberate slowing and acceleration, in order to approximate a natural speech tempo. Although this procedure does not guarantee uniform speech rate across participants, it reduces the likelihood of extreme realisations, such as unusually slow or unusually fast speech, that could independently affect production-based locality.

Another production-related source of variability is individual working memory load. Since working memory capacity and momentary load differ across participants and cannot be reliably controlled in our setting, this factor is best treated as an individual-level source of noise that affects the size of the production planning window and thus the likelihood of cross-word interaction. For this reason, we model speaker-specific differences statistically by including speaker as a random effect in the regression analysis. This allows us to quantify how much of the observed variability is attributable to between-speaker differences; as the results below show, this component is indeed substantial.

Having identified the factors that may influence the realisation of external sandhi, we can now control for the extra-grammatical ones in order to isolate the genuinely grammatical sources of variability. This is the goal of the production experiment presented in the following section.

4.5 Production experiment on pre-sonorant voicing

The production experiment was designed to test the influence of syntactic locality on the application of two external sandhi processes at the word boundary between a full word and a following (en)clitic: pre-sonorant voicing and final devoicing. The clitic position was occupied by two dative forms *mi* 'I.DAT' and *mu* 'he.DAT'. These clitics begin with the lexical sonorant /m/, which makes them suitable for investigating both pre-sonorant voicing and final devoicing.

The primary goal of the experiment was to determine the extent to which these two processes apply across different syntactic configurations, specifically comparing more local (verbal) and less local (adverbial) configurations. In all tested sentences, the clitics *mi* and *mu* appeared immediately after a single full word. This restriction formed part of the experimental control of extra-grammatical influences related to production planning (i.e., the factor of duration). The experimental design, therefore, aimed to minimise production-based variability in order to isolate the effect of syntactic locality as a grammatical conditioning factor.

The following subsections outline the experimental design, including the hypothesis, description of stimuli and participants, the data collection and annotation procedures, and the analytical results, followed by a discussion.

4.5.1 Experimental hypothesis

The experiment tests whether Czech dative pronominal clitics *mi* and *mu* behave uniformly across syntactic configurations, as expected under the standard characterisation of Czech pronominal clitics as position-selecting and encliticising.

On this view, the clitics are consistently second-position elements and are expected to encliticize in that position; their phonological behaviour should therefore not systematically depend on what happens to occupy clause-initial position. At the same time, the introduction motivates a competing possibility: even with an identical surface placement (and even with similar surface forms), different syntactic configurations may instantiate different degrees of syntactic deficiency and locality. If phonological processes at the clitic boundary reflect syntactic locality, then the same clitics may display different external sandhi behaviour depending on whether they occur adjacent to a structurally related constituent (verb) or a structurally unrelated one (adverb).

(17) a. <u>Null hypothesis</u> (∼ uniform behaviour of position-selecting clitics):
No systematic difference between the two configurations for either process; any variability should not be attributable to grammatical factors.

b. <u>Alternative hypothesis</u> (∼ locality-sensitive behaviour of position-selecting clitics):
Pre-sonorant voicing will be more likely when *mi* (or *mu*) follows a verbal constituent (local configuration) than when it follows an adverbial constituent (less local configuration). Final devoicing will show the complementary pattern, being more likely in the less local adverbial configuration than in the local verbal configuration.

4.5.2 Stimuli

In our experiment, we include three types of items: target stimuli, control stimuli, and fillers. The fillers consist of unrelated sentences that are irrelevant for the purposes of the current study and are therefore not listed here.[13]

Both target and control stimuli consist of monosyllabic or bisyllabic full words (w1) followed by an enclitic (w2). Two stimulus sets are prepared: one targeting pre-sonorant voicing and the other targeting final devoicing. In what follows, we describe the structure of the target and control stimuli in each set. The complete experimental sentences are provided in the Appendix.

The experimental items for pre-sonorant voicing are summarised in Table 4.14 below. The target stimuli are structured to test the influence of syntactic locality on pre-sonorant voicing. They are organised according to two syntactic locality conditions, expected to be more local (verb # CL) and expected to be less local (adverb ## CL), and according to the quality of the target-final fricative obstruent /s x ʃ/.

Table 4.14: Pre-sonorant voicing: set of experimental stimuli

	syntactic locality	final phoneme	target/control stimuli	
target items	expected to be more local (verb # CL)	/ s /	**přines mi** bring.IMP.2SG I.DAT	**přines mu** bring.IMP.2SG he.DAT
		/ x /	**pích mi** poke.PST.3SG I.DAT	**pích mu** poke.PST.3SG he.DAT
	expected to be less local (adverb ## CL)	/ s /	**dnes mi** today IDAT	**dnes mu** today he.DAT
control items	more local (QUANT # QUANT)	/ ʃ /	**příliš mnoho** too many	
	less local (SUBJ ## OBJ)	/ s /	**přednes nás** delivery we.ACC	

[13]Before selecting the target consonants for testing pre-sonorant voicing and final devoicing, it is necessary to consider their phonetic stability in word-final position. Specifically, we assess whether the segments in question are consistently realised across word boundaries or whether their word-final position leads to alternations or reductions that could obscure the realisation of either process. The decisive factor in this selection is the quality and stability of the acoustic signal at word boundaries, particularly with respect to voicing. Our earlier experiments have shown that word-final plosives, especially in tightly local configurations, frequently undergo partial or complete reduction. In such cases, the closure and release may be weakened or entirely absent, making the segment difficult or impossible to detect in the recording and thus preventing reliable assessment of both pre-sonorant voicing and final devoicing. The higher the degree of locality between the two words, the more likely such reductions become, in a manner comparable to cluster reduction effects (Bailey, 2019). To avoid this problem, we select segments that preserve their integrity at word boundaries. Fricatives meet this criterion better than plosives: fricative noise is consistently present and clearly segmentable, which increases the likelihood of obtaining analysable data without substantial data loss.

In the more local condition, w1 is a verb (*přines* or *pích*) and w2 is a pronominal enclitic (*mi* or *mu*). In the less local condition, w1 is an adverb (*dnes*) and w2 is a pronominal enclitic (*mi* or *mu*).[14]

The control stimuli consist of two additional configurations: a syntactically more local boundary within a complex quantifier phrase (*příliš mnoho*) and a syntactically less local subject–object boundary (*přednes nás*).

We now turn to the stimuli used for final devoicing, which are summarised in Table 4.15. This second set of stimuli is designed to test final devoicing under the same structural conditions as the pre-sonorant voicing set. Comparably, the target stimuli vary with respect to syntactic locality and the quality of the final obstruent.

Table 4.15: Final devoicing: set of experimental stimuli

	syntactic locality	final phoneme	target/control stimuli	
target items	expected to be more local (verb # CL)	/ z /	**pověz mi** tell.IMP.2SG I.DAT	**pověz mu** tell.IMP.2SG he.DAT
		/ ɦ /	**přistřih mi** clip.PST.3SG I.DAT	**přistřih mu** clip.PST.3SG he.DAT
	expected to be less local (adverb ## CL)	/ z /	**snáz mi** more.easily I.DAT	**snáz mu** more.easily he.DAT
control items	more local (adverb # verb)	/ d /	**snad neumřeme** perhaps not.die.PRES.1PL	
	less local (noun ## post-attribute)	/ z /	**výraz mi** expression I.DAT	**výraz mu** expression he.DAT

The segments /z/ and /ɦ/ are selected as the voiced counterparts of /s/ and /x/, which are used in the pre-sonorant voicing stimuli. This parallel design enables a direct comparison of final devoicing and pre-sonorant voicing under matched phonological and syntactic conditions. In the condition which is expected to be more local, w1 is a verb ending in /z/ or /ɦ/ (*pověz* or *přistřih*) and w2 is a pronominal enclitic (*mi* or *mu*). In the condition which is expected to be less local, w1 is an adverb ending in these segments (*snáz*) and w2 is a pronominal enclitic (*mi* or *mu*).

The control stimuli again represent instances of more local and less local configurations. In the more local configuration, w1 is an adverb (*snad*) followed by a verb (*neumřeme*). In the less local configuration, w1 is a noun (*výraz*) and w2 is a non-clitic form of the pronoun (*mi* or *mu*), forming a noun–post-attribute structure.

[14] Table 4.14 also shows that there are twice as many verb # CL pairs as adverb ## CL pairs. This imbalance is retained deliberately: while there is a limited number of adverbs ending in voiceless obstruents, Czech provides a rich inventory of verbs whose morphological forms end in such segments (e.g., imperatives or reduced *l*-participles forming the past tense). We have chosen to include this wider variety of verbal forms in order to mitigate potential lexical biases and reduce the influence of individual lexical items on the overall results.

4.5.3 Participants

The participants who provided the recordings were either students or faculty members from the Faculty of Arts at Masaryk University. The students took part in the experiment as part of the voluntary course Linguistic Experiment and received course credit for their participation. The faulty members participated on a voluntary basis.

The participants took part in elicitation sessions in which the target and control stimuli (Tables 4.14 and 4.15) were randomly interspersed with filler sentences. They were instructed to read all items as naturally as possible, at their own comfortable pace, as if addressing someone familiar. If participants were not satisfied with their first attempt at reading a sentence, they might repeat it until satisfied; for the analysis, the final attempt is used.

A total of 41 participants completed the experiment; however, one speaker has been excluded from the dataset due to consistent disfluencies leading to unnatural pauses within sentences. Both male and female speakers are included, though female participants are more numerous, reflecting the general demographics of the student population at the faculty.[15]

4.5.4 Collected data and annotation

The collected recordings have been subsequently processed by annotators trained in the analysis of linguistic speech data. The processing involves segmentation in *Audacity* and manual phonetic annotation based on a combination of auditory evaluation and visual inspection of spectrograms in *Praat* (Boersma and Weenink, 2026). The phonetic annotation followed the guidelines for Czech described in Machač and Skarnitzl (2009).

To detect both pre-sonorant voicing and final devoicing, we use the *Praat* function *Voicing: Fraction of locally unvoiced frames*, which calculates the proportion of unvoiced frames within a selected interval. The pitch range in *Praat* is set to 30–800 Hz, and these settings are used for the calculation of voicing values.

Rather than analysing only the word-final consonant, we examined the entire boundary cluster, consisting of the w1-final fricative and the w2-initial nasal /m/. This decision is motivated by the frequent coarticulation of the two consonants observed at this boundary, which make precise and consistent segmentation between the fricative and the nasal potentially subjective. Analysing the entire cluster reduces the risk of annotation bias resulting from arbitrary boundary placement.

On the basis of this metric, we adopt a binary classification. A value of 0 is assigned in cases where the cluster is fully voiced, corresponding to pre-sonorant voicing. Any value above 0 indicates the presence of voicelessness. Given the variability of this external sandhi process, we adopt a conservative criterion: only complete voicing assimilation is classified as an instance of pre-sonorant voicing. Any residual voicelessness is interpreted as the absence of the process. Both scenarios are illustrated in Figures 4.1 and 4.2. Figure 4.1 shows a case of pre-sonorant voicing, where the sonorant /m/ of the enclitic is preceded by a fully voiced consonant; the cluster is annot-

[15]The sample included speakers from both Moravian and Bohemian regions. However, since the experiment is already controlled for multiple variables, the inclusion of dialect as an additional factor proved statistically unreliable due to the limited sample size. We leave this question open for future research.

ated with a value of 0. Figure 4.2 illustrates a case where voicelessness persists in the cluster at the enclitic boundary, resulting in a value greater than 0.

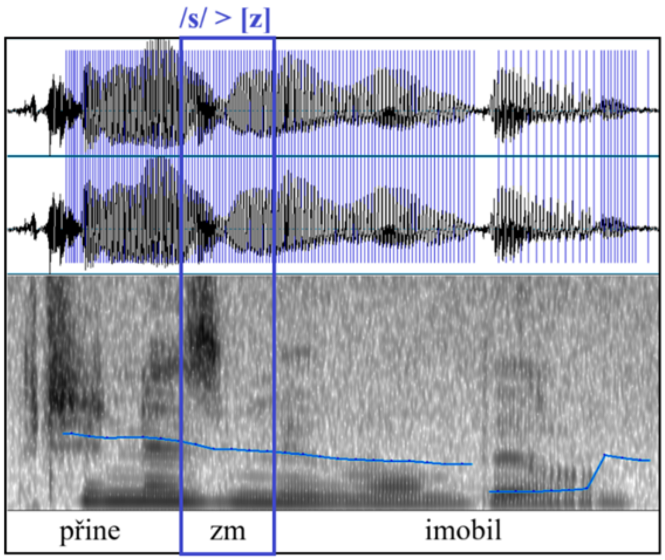

Figure 4.1: Spectrogram illustrating pre-sonorant voicing of w1-final obstruent: /s/ > [z]

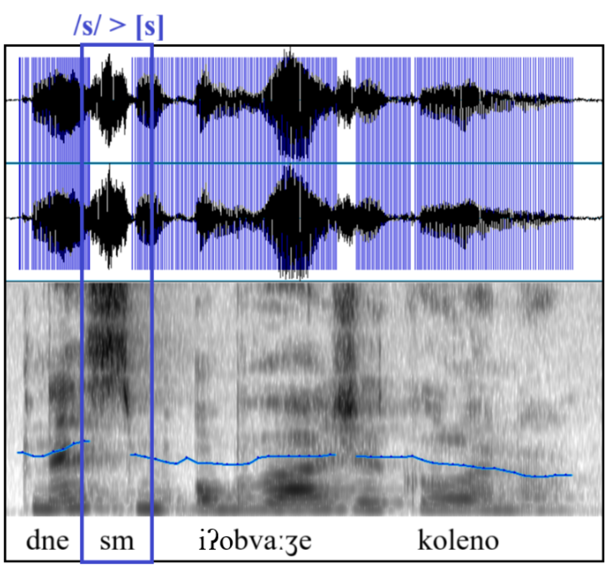

Figure 4.2: Spectrogram illustrating the absence of pre-sonorant voicing: /s/ > [s]

The key difference between the two spectrograms lies in the pitch contour, which is visualised by the horizontal curve. In Figure 4.2, this curve is interrupted within the boundary cluster, which we interpret as an indication of voicelessness and thus the absence of pre-sonorant voicing. In contrast, in Figure 4.1, the curve remains continuous across the cluster, which we take as evidence of full voicing and the application of pre-sonorant voicing. This visualisation of the pitch contour is based on the *Praat* function *Voicing: Fraction of locally unvoiced frames*. The consonant cluster either shows full voicing (fraction of locally unvoiced frames within the cluster is 0) or partial voicing (fraction of locally unvoiced frames within the cluster is bigger than 0). In the oscilogram, the absence of voicing is visualised by the absence of the vertical lines corresponding to the estimation of the glottis vibration.

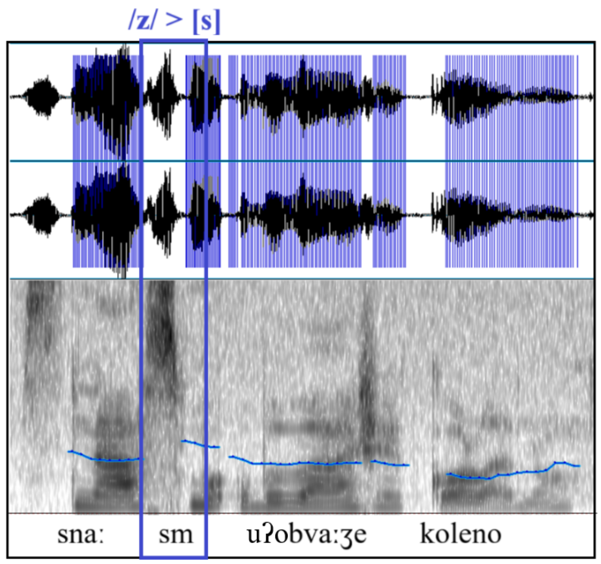

Figure 4.3: Spectrogram illustrating final devoicing of w1-final obstruent: /z/ > [s]

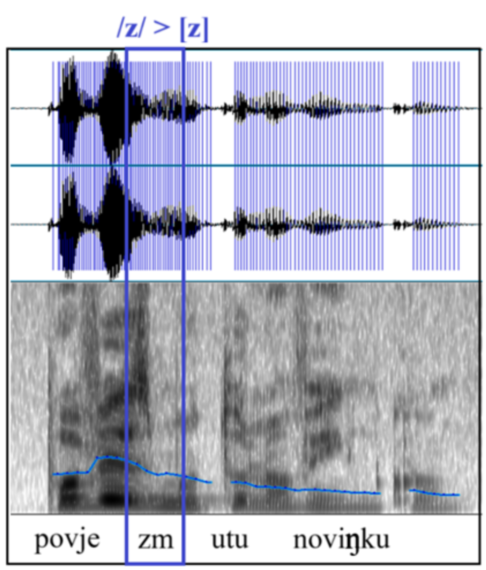

Figure 4.4: Spectrogram illustrating the absence of final devoicing: /z/ > [z]

It is worth noting, that the measured cluster never reaches 100% in the fraction of locally unvoiced frames, since its second part is always the phonetically voiced sonorant /m/.

For word-final devoicing, any value greater than 0 indicates that the process has applied: the lexically voiced w1-final consonant is realised as voiceless. This scenario is illustrated in Figure 4.3. Conversely, a value of 0 indicates that no devoicing has occurred, meaning that the consonant is produced as fully voiced; this is shown in Figure 4.4. As with pre-sonorant voicing, we apply a strict binary classification: final devoicing is considered absent only when no voicelessness is detectable in the measured segment (fraction of locally unvoiced frames within the cluster is 0).

These phonetically annotated data are evaluated using statistical tests, specifically a mixed-effects logistic regression model with syntactic locality as a fixed effect and speaker as a random effect, the results of which are summarized in the following subsections.

4.5.5 Results for pre-sonorant voicing at the clitic boundary

We first report the results for pre-sonorant voicing.[16]Table 4.16 summarises the absolute number of fully voiced clusters (measured value = 0) and clusters exhibiting any degree of voicelessness (measured value > 0) across the two experimental conditions. In the syntactically more local configurations, where w1 was a verb and w2 was the enclitic *mi* or *mu*, the boundary cluster remained voiceless in 120 cases and showed full pre-sonorant voicing in 28 cases. In the less local configuration, where w1 was an adverb, and w2 was *mi* or *mu*, only 2 responses were fully voiced, while 73 exhibited voicelessness.

Table 4.16: Pre-sonorant voicing in target items: token counts by condition[17]

syntactic locality	*application of pre-sonorant voicing*	*absence of pre-sonorant voicing*
expected to be more local (verb # CL)	28 (18.9%)	120 (81.9%)
expected to be less local (adverb ## CL)	2 (2.7%)	73 (97.3%)

The results in Table 4.16 suggest that voicing outcomes differ across the two locality conditions. To test this statistically, we conducted a χ-square test of independence. The analysis revealed a significant association between locality configuration and voicing outcome: $\chi^2(1, n = 223) = 11.469$, $p = 0.003$. We interpret this association as evidence that syntactic locality contributes to the likelihood of pre-sonorant voicing.

[16]The measurement outputs and annotated results from the production experiment are publicly available on Zenodo: Połomská and Ziková (2026). Audio recordings are not included due to GDPR constraints.

[17]The table reports 223 out of total 240 target stimuli. The discrepancy is due to 17 defective items, which are excluded from the analysis.

To address possible speaker variability, we fitted a mixed-effects logistic regression model to estimate how strongly locality configuration predicts the probability of a fully voiced realisation, while controlling for inter-speaker variation (see Table 4.17).

Table 4.17: Results of a mixed-effects logistic regression examining the effect of syntactic locality on pre-sonorant voicing (adverb compared to verb), with speaker variability controlled for

Predictors	*Odds Ratios*	*Std. Error*	*Conf. Interval*	*p*
(Intercept)	0.00	0.00	0.00 – Inf	**<0.001**
w1 [verb]	45.23	0.05	0.00 – Inf	**<0.001**
Random Effects				
σ^2	3.29			
$\tau_{00speaker}$	18.63			
ICC	0.85			
$N_{speaker}$	40			
Observations	240			
Marginal R^2 / Conditional R^2	0.129 / 0.869			

To evaluate the effect of syntactic locality while controlling for individual variation, we fitted a mixed-effects logistic regression model with speaker as a random intercept. The random-effect variance was substantial ($\tau^2 = 18.63$ vs $\sigma^2 = 3.29$), yielding an intra-class correlation of 0.85. This indicates that a large proportion of the variance is attributable to between-speaker differences, with speakers differing markedly in their likelihood of producing fully voiced realisations.
Fixed effects alone account for a modest share of the variance (marginal $R^2 = 0.129$), whereas the full model, including speaker-level variation, explains most of it (conditional $R^2 = 0.869$). Despite this strong inter-speaker variability, the effect of w1 type remains statistically robust. Relative to the adverb condition, the verb condition shows a markedly higher likelihood of pre-sonorant voicing (OR = 45.23, $p < 0.001$). This confirms a strong locality asymmetry at the w1–w2 boundary in the production data.

Comparing the target items with the control items, the pattern is consistent with the intended locality contrast. The adverb ## CL target items behave essentially like the less local control SUB ## OBJ: in both cases, pre-sonorant voicing is largely absent (adverb ## CL: 2.7% application; SUB ## OBJ: 7.5% application of pre-sonorant voicing). By contrast, the verb # CL target items pattern closer to the more local control items QUANT # QUANT, in that they show a clearly higher rate of application of external sandhi process than the less local configurations. At the same time, they do not match the more local control items quantitatively (verb # CL: 18.9% VS QUANT # QUANT: 82.1%), indicating that verb–enclitic sequences pattern as more local relative to the adverb condition, but still show substantial variability.

Table 4.18: Pre-sonorant voicing in control items: token counts by condition[18]

syntactic locality	*application of pre-sonorant voicing*	*absence of pre-sonorant voicing*
more local (QUANT # QUANT)	32 (82.1%)	7 (17.9%)
less local (SUBJ ## OBJ)	3 (7.5%)	37 (92.5%)

For the finding on pre-sonorant voicing to carry full weight, the experiment will of course need to be replicated with a substantially larger set of tested items; nevertheless, it is already clear that different types of first-position constituents instantiate different locality configurations at the enclitic boundary, and that designing and running an experiment of this kind was methodologically well motivated.

4.5.6 Discussion of pre-sonorant voicing results

The standard characterisation of Czech pronominal clitics as position-selecting (Wackernagel) provides a clear baseline for interpreting our results. On this view, the placement of *mi* and *mu* is determined by the second-position requirement in the clause, not by the identity or category of w1. The clitics are therefore not expected to be obligatorily adjacent to any particular constituent (e.g. as verb-adjacent clitics in Romance languages). In other words, second-position clitics should not behave differently depending on whether w1 is a verb or an adverb. Their encliticisation process should therefore be uniform across the two conditions.

Yet the results of our experiment show that the two conditions (w1 = verb and w1 = adverb) yield different values for pre-sonorant voicing. This contrast shows that the two configurations are not phonologically equivalent and supports the conclusion that they instantiate two distinct syntactic configurations, reflected in two distinct phonological patterns at the enclitic boundary. Different phonological behaviour corresponds to different phonological locality, which in turn corresponds to different syntactic locality. The results therefore show that Czech enclitics are sensitive to syntactic locality in their phonological behaviour: pre-sonorant voicing is more likely in the more local (verb # CL) configuration than in the less local (adverb ## CL) configuration.

The overall rate of pre-sonorant voicing in the experiment is lower than what might be expected under categorical accounts, but this is consistent with the corpus results, which already show that the process is variable even before the enclitics *mi* and *mu*. In addition, our coding deliberately adopts a conservative criterion, counting only complete voicing assimilation (measured value = 0) as an instance of pre-sonorant voicing. Under this strict classification, the crucial point is not the absolute rate of application, but the relative difference between the two syntactic conditions.

[18] One more local control item is defective and thus excluded from the analysis, the total number of control items in the more local condition is thus 39.

In sum, the experimental data provide evidence that Czech pronominal clitics, despite being position-selecting in their distribution and despite being traditionally considered enclitic, show differential external sandhi behaviour depending on the syntactic locality of the w1–w2 configuration. This supports the view that locality effects described for other external sandhi patterns can extend to Czech enclitics as well, a possibility that has not been explicitly established in the existing literature. If the two conditions differ in syntactic locality, they also differ in syntactic configuration, and this should be taken into account in syntactic analyses.

We have now seen that the results for pre-sonorant voicing at the clitic boundary point to a clear effect of syntactic locality. In the following section, we turn to the complementary process of final devoicing to examine whether it reveals the same pattern.

4.5.7 Results for final devoicing at the clitic boundary

We now turn to the results for final devoicing, measured at a similar w1–w2 boundary as in the pre-sonorant voicing analysis. Table 4.19 summarises the distribution of the final devoicing data. In the more local configuration, where w1 is a verb, and w2 is the enclitic (*mi* or *mu*), final devoicing is attested in 112 cases (value > 0), while 33 tokens remain fully voiced (value = 0). In the less local configuration, where w1 is an adverb and w2 is the enclitic, 65 tokens show final devoicing, while only 6 remain voiced.

Table 4.19: Final devoicing: token counts by condition[19]

syntactic locality	*application of final devoicing*	*absence of final devoicing*
expected to be more local (verb # CL)	112 (77.2%)	33 (22.8%)
expected to be less local (adverb ## CL)	65 (91.5%)	6 (8.5%)

At first glance, the results for final devoicing appear straightforward. To test the effect statistically, we conducted a χ-square test of independence. The analysis confirms that the distribution of devoicing differs significantly between the more local and less local configurations: $\chi^2(1, n = 216) = 6.757, p = 0.034$.

As with pre-sonorant voicing, it is necessary to check whether this pattern holds across speakers or is driven by a subset of participants. To evaluate the effect more robustly, we fitted a mixed-effects logistic regression model predicting the binary presence of voicing in the word-final obstruent (see Table 4.20). The model included a fixed effect of w1 type (verb) and random intercepts for speaker. The speaker-level variance ($\tau^2 = 3.02$) was substantial relative to residual variance ($\sigma^2 = 3.29$), yielding an intra-class correlation (ICC) of 0.48. This suggests that nearly

[19] The table reports 216 out of a total of 240 target stimuli. The discrepancy is due to 24 defective tokens, which are excluded from the analysis.

half of the variance in voicing outcomes is attributable to between-speaker differences, indicating that speakers differ substantially in how consistently they apply final devoicing across w1 types. Given that we are testing an external sandhi process, such inter-speaker variability is expected.

Table 4.20: Results of a mixed-effects logistic regression examining the effect of w1 locality on final devoicing (adverb compared to verb), with speaker variability controlled for

Predictors	*Odds Ratios*	*Std. Error*	*Conf. Interval*	*p*
(Intercept)	0.03	0.02	0.00 – Inf	**<0.001**
w1 [verb]	4.94	2.79	0.00 – Inf	**0.005**
Random Effects				
σ^2	3.29			
$\tau_{00 speaker}$	3.02			
ICC	0.48			
$N_{speaker}$	40			
Observations	240			
Marginal R^2 / Conditional R^2	0.083 / 0.522			

While the fixed effect of w1 type alone accounts for only a small portion of the variance in the data (marginal $R^2 = 0.083$), the full model, including between-speaker variation, explains more than half of the variability in devoicing outcomes (conditional $R^2 = 0.522$). This indicates that, although w1 type plays a clear and statistically significant role, individual speakers vary considerably in how they apply final devoicing. Crucially, despite this variability, the effect of w1 type remains robust, providing evidence against the view that enclitic phonology is uniformly insensitive to syntactic configuration. Our data show the opposite: final devoicing occurs significantly more often when the enclitic is preceded by an adverb than by a verb (OR = 4.94, $p = 0.005$). This asymmetry is crucial, as it suggests that clitics do not form a single phonological word with adverbs in the same way they do with verbs. The presence of word-final devoicing indicates a phonological boundary between the adverb and the clitic, challenging the claim that Czech enclitics are categorically insensitive to syntactic locality.

Furthermore, final devoicing in the target items aligns with the control items in the expected direction, given that more final devoicing corresponds to a less local configuration, see Table 4.21.

Table 4.21: Final devoicing in control items: token counts by condition[20]

syntactic locality	*application of final devoicing*	*absence of final devoicing*
more local (adverb # verb)	12 (32.4%)	25 (67.6%)
less local (noun ## post-attribute)	71 (95.9%)	3 (4.1%)

The adverb ## CL target items behave like the less local control items (noun ## post-attribute): both show near-categorical devoicing (adverb ## CL: 91.5%; noun ## post-attribute: 95.9%). By contrast, the verb # CL target items show less devoicing (77.2%). Although this rate is not as low as in the more local control (adverb # verb), where devoicing is much less frequent (32.4%), it is still clearly lower than in the adverb ## CL condition. This means that verb # CL does not behave like our most local baseline, but it does differ systematically from adverb ## CL, indicating that the two target configurations involve different locality conditions at the clitic boundary.

4.5.8 Discussion of final devoicing results

The standard characterisation of Czech pronominal clitics as position-selecting (Wackernagel) and encliticising provides the relevant baseline for interpreting the present results. If this traditional view is correct, no difference should be expected in the phonological behaviour of the clitics across the two conditions examined here. Their placement is determined by the second-position requirement in the clause, not by the identity or category of w1, and their encliticisation should therefore proceed uniformly whether w1 is a verb or an adverb. Under this expectation, the w1–w2 boundary should not display different phonological behaviour depending on whether the clitic follows a verb or an adverb.

The results, however, do not support this traditional expectation. As shown in the preceding section, pre-sonorant voicing is more likely in the more local (verb # CL) configuration than in the less local (adverb ## CL) configuration. If the same contrast reflects a genuine asymmetry in locality at the clitic boundary, it should also be detectable in the opposite direction through the complementary process of final devoicing.

Final devoicing provides a complementary perspective on the pre-sonorant voicing results, because it targets the same w1–w2 boundary but yields the opposite diagnostic: whereas pre-sonorant voicing signals greater phonological locality, final devoicing signals a stronger boundary and reduced locality. For final devoicing, the expected pattern is therefore the reverse of pre-sonorant voicing: devoicing should be favoured in the less local (adverb ## CL) configuration and disfavoured in the more local (verb # CL) configuration. The results match this prediction. Final devoicing is

[20] Three more local control items are defective and thus excluded from the analysis; the total number of control items in the more local condition is thus 37. Nine fewer local control items are defective and thus excluded from the analysis; the total number of control items in the less local condition is thus 74.

significantly more frequent when w1 is an adverb than when w1 is a verb. Since w2 is held constant and surface linearisation is comparable in both conditions, the contrast cannot be reduced to the clitic itself or to any other segmental factor. Rather, it indicates that the w1–w2 boundary is not phonologically uniform across the two syntactic configurations: after adverbs, it behaves as less local, whereas after verbs, it behaves as more local.

Crucially, this pattern aligns with the pre-sonorant voicing results. The two complementary processes converge on the same locality asymmetry from opposite directions: in the adverb condition, pre-sonorant voicing is strongly reduced and final devoicing is favoured, both patterns consistent with a stronger boundary between w1 and w2. In the verb condition, by contrast, the boundary behaves as weaker, yielding less final devoicing and a higher likelihood of pre-sonorant voicing.

In sum, the final devoicing data reinforce the main conclusion drawn from pre-sonorant voicing: phonological behaviour at the w1–w2 boundary reflects the syntactic locality of the configuration, even at the clitic boundary in a language where enclitics are generally assumed to behave uniformly.

4.6 Conclusion

This study has shown that Czech pronominal clitics in second position do not display uniform phonological behaviour. Once the null hypothesis of uniform behaviour is rejected, the results are interpreted in terms of the alternative hypothesis based on syntactic locality. Using a controlled production experiment, we tested two complementary phonological processes, pre-sonorant voicing and final devoicing, in two syntactic configurations: clitics following verbs (more local) and clitics following adverbs (less local). The results converge on the same asymmetry. In the more local verbal configuration, pre-sonorant voicing is attested but variable, and final devoicing is comparatively disfavoured. In the less local adverbial configuration, pre-sonorant voicing is strongly reduced, while final devoicing is favoured. The experimental design also controlled for major production-planning factors, reducing extragrammatical variability at the tested boundary. Importantly, these effects remain robust even when substantial inter-speaker variability, expected for external sandhi, is taken into account.

These findings bear directly on the standard view of Czech pronominal clitics as position-selecting (Wackernagel) enclitics. Since these clitics occupy the same clausal position in both configurations, the observed contrast cannot be attributed to differences in clitic distribution alone. Under this view, uniform distribution would also be expected to correlate with uniform phonological behaviour. The results show, however, that phonological behaviour at the clitic boundary is conditioned by syntactic configuration: the same clitics pattern differently depending on whether w1 is a verb or an adverb. This, in turn, implies that clauses with clitics in which w1 is a verb differ structurally from those in which w1 is an adverb in a way that is relevant for phonological integration.

More broadly, the study also revives a method of detecting syntactic locality through the observation of phonological processes at word boundaries, specifically external sandhi and final devoicing. The parallel behaviour of these two complementary processes provides a consistent diagnostic of locality at the clitic boundary. The less local configuration patterns as a stronger boundary, suppressing pre-sonorant voicing and promoting final devoicing, whereas the more local configuration patterns as a weaker boundary, permitting greater phonological interaction and therefore yielding more variability. In this way, the results extend locality-sensitive sandhi effects to Czech (en)clitics, a domain in which such asymmetries have not been clearly established in the existing literature.

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Appendix

Table 4.22: Queries for pre-sonorant voicing before any word starting with the phoneme /m/ in *Ortofon v3*

voiceless phoneme	query for lexically voiceless phoneme pronounced as voiceless	query for lexically voiceless phoneme pronounced as voiced
/ x /	[fon=".*ch" & word=".*ch"][word="m.*"]	[fon=".*ɣ" & word=".*ch"][word="m.*"]
/s/	[fon=".*s" & word=".*s"][word="m.*"]	[fon=".*z" & word=".*s"][word="m.*"]
/ ʃ /	[fon=".*š" & word=".*š"][word="m.*"]	[fon=".*ž" & word=".*š"][word="m.*"]
/ f /	[fon=".*f" & word=".*f"][word="m.*"]	[fon=".*v" & word=".*f"][word="m.*"]
/ k /	[fon=".*k" & word=".*k"][word="m.*"]	[fon=".*g" & word=".*k"][word="m.*"]
/ t /	[fon=".*t" & word=".*t"][word="m.*"]	[fon=".*d" & word=".*t"][word="m.*"]
/ c /	[fon=".*ť" & word=".*ť"][word="m.*"]	[fon=".*ď" & word=".*ť"][word="m.*"]
/ p /	[fon=".*p" & word=".*p"][word="m.*"]	[fon=".*b" & word=".*p"][word="m.*"]

Table 4.23: Pre-sonorant voicing before any word starting with the phoneme /m/ in *Ortofon v3*

voiceless phoneme	pronounced as voiceless (⇒ no pre-sonorant voicing)		pronounced as voiced (⇒ pre-sonorant voicing)		tot.
	abs. freq.	rel. freq. (%)	abs. freq.	rel. freq. (%)	
/ x /	628	76.3	**195**	**23.7**	823
/ s /	946	55.8	**749**	**44.2**	1695
/ ʃ /	773	65.8	**401**	**34.2**	1174
/ f /	6	66.7	**3**	**33.3**	9
/ k /	2902	64.9	**1569**	**35.1**	4471
/ t /	1003	67.9	**475**	**32.1**	1478
/ c /	136	61.3	**86**	**38.7**	222
/ p /	26	61.9	**16**	**38.1**	42
tot.	6420	64.8	**3494**	**35.2**	9914

Table 4.24: Queries for final devoicing before any word starting with the phoneme /m/ in *Ortofon v3*

voiceless phoneme	query for lexically voiceless phoneme pronounced as voiceless	query for lexically voiceless phoneme pronounced as voiced
/ ɦ /	[fon=".*ch" & word=".*h" & word!=".*ch"] [word="m.*"]	[fon=".*h" & word=".*h" & word!=".*ch"] [word="m.*"] MINUS [fon=".*ch" & word=".*h" & word!=".*ch"][word="m.*"]
/ z /	[fon=".*s" & word=".*z"][word="m.*"]	[fon=".*z" & word=".*z"][word="m.*"]
/ ʒ /	[fon=".*š" & word=".*ž"][word="m.*"]	[fon=".*ž" & word=".*ž"][word="m.*"]
/ v /	[fon=".*f" & word=".*v"][word="m.*"]	[fon=".*v" & word=".*v"][word="m.*"]
/ g /	[fon=".*k" & word=".*g"][word="m.*"]	[fon=".*g" & word=".*g"][word="m.*"]
/ d /	[fon=".*t" & word=".*d"][word="m.*"]	[fon=".*d" & word=".*d"][word="m.*"]
/ ɟ /	[fon=".*ť" & word=".*ď"][word="m.*"]	[fon=".*ď" & word=".*ď"][word="m.*"]
/ b /	[fon=".*p" & word=".*b "][word="m.*"]	[fon=".*b" & word=".*b"][word="m.*"]

Table 4.25: Final devoicing before any word starting with the phoneme /m/ in *Ortofon v3*

voiced phoneme	pronounced as voiceless (⇒ final devoicing)		pronounced as voiced (⇒ no final devoicing)		tot.
	abs. freq.	rel. freq. (%)	abs. freq.	rel. freq. (%)	
/ ɦ /	**24**	**82.8**	5	17.2	29
/ z /	**26**	**11.1**	208	88.9	234
/ ʒ /	**1438**	**65.9**	745	34.1	2183
/ v /	**34**	**10.4**	293	89.6	327
/ g /	**4**	**57.1**	3	42.9	7
/ d /	**160**	**42.0**	221	58.0	381
/ ɟ /	**234**	**51.7**	219	48.3	453
/ b /	**8**	**88.9**	1	11.1	9
tot.	**1928**	**53.2**	1695	46.8	3623

Table 4.26: Queries for pre-sonorant voicing before *mi* 'I.DAT' or *mu* 'he.DAT' in *Ortofon v3*

voiceless phoneme	query for lexically voiceless phoneme pronounced as voiceless	query for lexically voiceless phoneme pronounced as voiced
/ x /	[fon=".*ch" & word=".*ch"][word="m[ui]"]	[fon=".*ɣ" & word=".*ch"][word="m[ui]"]
/ s /	[fon=".*s" & word=".*s"][word="m[ui]"]	[fon=".*z" & word=".*s"][word="m[ui]"]
/ ʃ /	[fon=".*š" & word=".*š"][word="m[ui]"]	[fon=".*ž" & word=".*š"][word="m[ui]"]
/ f /	[fon=".*f" & word=".*f"][word="m[ui]"]	[fon=".*v" & word=".*f"][word="m[ui]"]
/ k /	[fon=".*k" & word=".*k"][word="m[ui]"]	[fon=".*g" & word=".*k"][word="m[ui]"]
/ t /	[fon=".*t" & word=".*t"][word="m[ui]"]	[fon=".*d" & word=".*t"][word="m[ui]"]
/ c /	[fon=".*ť" & word=".*ť"][word="m[ui]"]	[fon=".*ď" & word=".*ť"][word="m[ui]"]
/ p /	[fon=".*p" & word=".*p"][word="m[ui]"]	[fon=".*b" & word=".*p"][word="m[ui]"]

Table 4.27: Pre-sonorant voicing before enclitics *mi* 'I.DAT' and *mu* 'he.DAT' in *Ortofon v3*

voiceless phoneme	pronounced as voiceless (⇒ no pre-sonorant voicing)		pronounced as voiced (⇒ pre-sonorant voicing)		tot.
	abs. freq.	rel. freq. (%)	abs. freq.	rel. freq. (%)	
/ x /	60	73.2	**22**	**26.8**	82
/ s /	164	62.8	**97**	**37.2**	261
/ ʃ /	168	75.0	**56**	**25.0**	224
/ f /	0	0.0	**1**	**100.0**	1
/ k /	494	66.0	**254**	**34.0**	748
/ t /	56	71.8	**22**	**28.2**	78
/ c /	37	54.4	**31**	**45.6**	68
/ p /	6	85.7	**1**	**14.3**	7
tot.	976	66.8	**484**	**33.2**	1460

Table 4.28: Queries for final devoicing before any word starting with the phoneme /m/ in *Ortofon v3*

voiceless phoneme	query for lexically voiceless phoneme pronounced as voiceless	query for lexically voiceless phoneme pronounced as voiced
/ ɦ /	[fon=".*ch" & word=".*h" & word!=".*ch"] [word="m[ui]"]	[fon=".*h" & word=".*h" & word!=".*ch"] [word="m[ui]"] MINUS [fon=".*ch" & word=".*h" & word!=".*ch"][word="m.*"]
/ z /	[fon=".*s" & word=".*z"][word="m[ui]"]	[fon=".*z" & word=".*z"][word="m[ui]"]
/ ʒ /	[fon=".*š" & word=".*ž"][word="m[ui]"]	[fon=".*ž" & word=".*ž"][word="m[ui]"]
/ v /	[fon=".*f" & word=".*v"][word="m[ui]"]	[fon=".*v" & word=".*v"][word="m[ui]"]
/ g /	[fon=".*k" & word=".*g"][word="m[ui]"]	[fon=".*g" & word=".*g"][word="m[ui]"]
/ d /	[fon=".*t" & word=".*d"][word="m[ui]"]	[fon=".*d" & word=".*d"][word="m[ui]"]
/ ɟ /	[fon=".*ť" & word=".*ď"][word="m[ui]"]	[fon=".*ď" & word=".*ď"][word="m[ui]"]
/ b /	[fon=".*p" & word=".*b"][word="m[ui]"]	[fon=".*b" & word=".*b"][word="m[ui]"]

Table 4.29: Final devoicing before enclitics *mi* 'I.DAT' and *mu* 'he.DAT' in *Ortofon v3*

voiced phoneme	pronounced as voiceless (⇒ final devoicing)		pronounced as voiced (⇒ no final devoicing)		tot.
	abs. freq.	rel. freq. (%)	abs. freq.	rel. freq. (%)	
/ ɦ /	**3**	**100.0**	0	0.0	3
/ z /	**3**	**60.0**	2	40.0	5
/ ʒ /	**251**	**65.9**	130	34.1	381
/ v /	**1**	**12.5**	7	87.5	8
/ g /	**0**	**0.0**	0	0.0	0
/ d /	**20**	**52.6**	18	47.4	38
/ ɟ /	**46**	**43.4**	60	56.6	106
/ b /	**1**	**100**	0	0	1
tot.	**325**	**60.0**	217	40.0	542

Table 4.30: Stimuli for pre-sonorant voicing experiment

locality	final phoneme	target stimuli
expected to be more local (verb # CL)	/ s /	**Přines mi** mobil. bring.IMP.2SG I.DAT phone.ACC 'Bring me the phone.'
	/ s /	**Přines mu** to do kanceláře. bring.IMP.2SG he.DAT it.ACC to office.GEN 'Bring him it to the office.'
	/x/	**Pích mi** trn do ruky. poke.PST.3SG I.DAT thorn.ACC into hand.GEN 'He poked me a thorn into the hand.'
	/ x /	**Pích mu** trn do ruky. poke.PST.3SG he.DAT thorn.ACC into hand.GEN 'He poked him a thorn into the hand.'
expected to be less local (adverb ## CL)	/ s /	**Dnes mi** obváže koleno. today I.DAT bandage.PRES.3SG knee.ACC 'Today, he will bandage me the knee.'
	/ s /	**Dnes mu** obváže koleno. today he.DAT bandage.PRES.3SG knee.ACC 'Today, he will bandage him the knee.'
locality	**final phoneme**	**control stimuli**
more local (QUANT # QUANT)	/ ʃ /	Děláte **příliš mnoho** chyb. do.PRES.2PL too.many mistakes.GEN 'You are making too many mistakes.'
less local (SUBJ ## OBJ)	/ s /	Jeho strojený **přednes nás** překvapil. his affected delivery.NOM we.ACC surprise.PST.3SG 'His affected delivery surprised us.'

Table 4.31: Stimuli for final devoicing experiment

locality	final phoneme	target stimuli
expected to be more local (verb # CL)	/ z /	**Pověz mi** o tom víc. tell.IMP.2SG I.DAT about that.LOC more 'Tell me more about that.'
	/ z /	**Pověz mu** tu novinku. tell.IMP.2SG he.DAT that news.ACC 'Tell him the news.'
	/ ɦ /	**Přistřih mi** křidýlka. clip.PST.3SG I.DAT wings.ACC 'He clipped my wings.'
	/ ɦ /	**Přistřih mu** křidýlka. clip.PST.3SG he.DAT wings.ACC 'He clipped his wings.'
expected to be less local (adverb ## CL)	/ z /	**Snáz mi** obváže koleno. more.easily I.DAT bandage.PRES.3SG knee.ACC 'He will bandage my knee more easily.'
	/ z /	**Snáz mu** obváže koleno. more.easily he.DAT bandage.PRES.3SG knee.ACC 'He will bandage his knee more easily.'

locality	final phoneme	control stimuli
more local (adverb # verb)	/ d /	**Snad neumřeme** ještě dnes. perhaps not.die.PRES.1PL still today 'Hopefully we won't die today.'
less local (noun ## post-attribute)	/ z /	**Výraz mi** se do tohoto spojení nehodí. expression.NOM I.DAT REFL to this phrase not.fit.PRES.3SG 'The expression *mi* doesn't fit this phrase.'
	/ z /	**Výraz mu** se do tohoto spojení nehodí. expression.NOM he.DAT REFL to this phrase not.fit.PRES.3SG 'The expression *mu* doesn't fit this phrase.'